

• DEVOPS BASICS

What is CI/CD?

The automated path from a **git commit** to **production** — build, test & ship without doing it by hand. Let's break it down and wire one up with GitHub Actions.

• THE TWO HALVES

CI vs CD

CI

Continuous Integration

Every push is **automatically built & tested**, so broken code and bugs surface early — not in production.

CD

Continuous Delivery / Deployment

Once tests pass, the app is **released automatically** to staging or production — no manual, error-prone deploys.

• ANATOMY

Inside a pipeline



Trigger

A push or pull request kicks the pipeline off.



Build

Install dependencies and compile the app.



Test

Run automated tests — fail fast if something breaks.



Deploy

Ship the passing build to staging or production.

Now let's build one →

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• GITHUB ACTIONS

A basic pipeline

.github/workflows/ci.yml

ci.yml

```
name: CI
on: [push]           # run on every push

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
      - run: npm install
      - run: npm test
      - run: npm run build
```


• RECAP

That's CI/CD.

- ✓ CI — auto build & test on every push.
- ✓ CD — auto release once tests pass.
- ✓ GitHub Actions wires it up with one YAML file.
- ✓ Start small — a single workflow that installs, tests & builds.

💬 What does your CI/CD setup look like?